FURTHER MECHANICS / WORK, ENERGY AND POWER

Power and Driving Force

Mark scheme 12 questions · 107 marks

Questions

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	Total	**107**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$P = Fv$ with $F = R$: $200\,000 = 20R \Rightarrow R = \ldots$	**M1**	Uses $P = Fv$ with driving force equal to resistance at $v = 20$. Condone power left in kW.
	$R = 10\,000\,\text{N}$	**A1**	
		[2]	
1(b)	$\frac{60\,000}{8} - \text{“}10\,000\text{”} = 10\,000a \Rightarrow a = \ldots$	**M1**	Forms an equation of motion with driving force $\frac{P}{v}$, their resistance and mass 10 000. Condone sign slips and power in kW.
	$a = -0.25\,\text{m}\,\text{s}^{-2}$	**A1**	Or a deceleration of 0.25. A0 for $a = 0.25$ with no direction.
		[2]	
1(c)	$F = \frac{200\,000}{10} = 20\,000$	**B1**	
	$\text{“}20\,000\text{”} = \text{“}10\,000\text{”} + 10\,000 \times 9.8 \sin\theta$	**M1 A1**	**M1** Resolves along the slope with zero acceleration, using their driving force, their resistance and a weight component. Condone sign slips and sin/cos confusion. **A1** Correct equation, oe.
	$\sin\theta = \frac{5}{49} \Rightarrow \theta = 5.9°$	**A1**	awrt 5.9°, or awrt 0.102 radians.
		[4]	
Total for Question 1		**8**	

Question	Scheme	Marks	Notes
2	$3000g\sin\theta = 3000 \times 9.8 \times \frac{1}{49} = 600$	**B1**	
	$P = Fv$ $F =$ "600" $+ 300 + 60v \Rightarrow P = \ldots$	**M1**	Uses $P = Fv$ with driving force equal to resistance plus a weight component, at speed v or 15. Condone sign slips.
	$P = 15(600 + 300 + 60 \times 15)$	**A1**	Correct expression for the power at $v = 15$, oe.
	Power needed, $900v + 60v^2$, increases with v, so the greatest constant speed is at maximum power	**E1**	States that the engine works at maximum power at the greatest constant speed, with a reason such as the power needed increasing with v.
	$P_{\max} = 27\,000\,\text{W}$	**A1**	
	Total for Question 2	**5**	

Question	Scheme	Marks	Notes
3(i)	$P = Fv$ with $F = R$: $72\,000 = 18R \Rightarrow R = \ldots$	**M1**	Uses $P = Fv$ at $v = 18$ with the driving force equal to the resistance.
	$R = 4000\,\text{N}$	**A1**	
		[2]	
3(ii)	$\frac{72\,000}{12}$ − "4000" $= \ldots$ $[= 6000 - 4000]$	**M1**	Finds the total resistance from $P = Fv$ at $v = 12$ and subtracts their tug resistance.
	$2000\,\text{N}$	**A1**	
		[2]	
3(iii)(a)	F − "4000" − "2000" $= 4000 \times 0.25$ $\Rightarrow F = \ldots$ $[= 7000]$	**M1 A1**	**M1** Forms Newton's second law for the whole system, with driving force, both their resistances and the total mass. Condone sign slips. **A1** Correct equation, oe.
	$P = Fv =$ "7000" $\times 10$	**M1**	Uses $P = Fv$ with their driving force from Newton's second law and $v = 10$.
	$P = 70\,000\,\text{W}$ $(70\,\text{kW})$	**A1**	
		[4]	
3(iii)(b)	T − "2000" $= 2200 \times 0.25 \Rightarrow T = \ldots$	**M1**	Forms Newton's second law for the aircraft alone, or the tug alone with their driving force, including the tow bar force. Condone sign slips.
	$T = 2550\,\text{N}$	**A1**	
		[2]	
3(iv)(a)	Lower: no resistance below $6\,\text{m}\,\text{s}^{-1}$, so greater resultant force and acceleration there	**E1**	States lower.
	Models agree from 6 to $10\,\text{m}\,\text{s}^{-1}$, so total time is less	**E1**	States the models agree from 6 to 10, so the time saved below 6 makes the total time less.
		[2]	
3(iv)(b)	The same	**E1**	States that the greatest steady speed is the same.

Question 3 continued

Question	Scheme	Marks	Notes
	$12 > 6$, so same resistances and same maximum power at that speed	**E1**	Explains that the steady speed of 12 exceeds 6, so the original resistances and the same maximum power apply there.
		[2]	
	Total for Question 3	**14**	

Question	Scheme	Marks	Notes
4(a)	$P = Fv \Rightarrow 12\,000 = 15F \Rightarrow F = 800$	**B1**	
	"800" $= R + 1000 \times 9.8 \times \frac{1}{20}$	**M1 A1**	**M1** Resolves along the slope with zero acceleration, with their driving force, R and the weight component. Condone sign slips and sin/cos confusion.
	$R = 310$	**A1**	
		[4]	
4(b)	$F = \frac{12\,000}{7.5} = 1600$	**B1**	
	"1600" − "310" − 106 − $1600 \times 9.8 \times \frac{1}{20}$ $= 1600a$	**M1 A1**	**M1** Forms an equation of motion for the system, or for the van alone with T, with driving force, all resistances and weight components. Condone sign slips and sin/cos confusion.
	$a = 0.25$	**A1**	
	$T - 106 - 600 \times 9.8 \times \frac{1}{20}$ $= 600 \times$ "0.25"	**M1 A1**	**M1** Forms a second equation of motion with T, for the trailer or the van, with all its forces and mass times acceleration. Condone sign slips and sin/cos confusion.
	$T = 550$	**A1**	
		[7]	
Total for Question 4		**11**	

Question	Scheme	Marks	Notes
5	Maximum power at the greatest constant speed and for the greatest acceleration: $P = Fv \Rightarrow F = \frac{64\,000}{v}$	**E1**	States maximum power at $v = 20$ and at $v = 16$, each with a reason. Both needed.
	$\frac{64\,000}{20} = (200 + 20k) + 2000g\sin\alpha$	**M1 A1**	**M1** Resolves along the slope with zero acceleration, using driving force $\frac{64\,000}{20}$, resistance at $v = 20$ and a weight component. Condone sign slips.
	$k = 101$	**A1**	
	$\frac{64\,000}{16} - (200 + 16\text{“}k\text{”}) - 2000g\sin\alpha = 2000a$	**M1 A1**	**M1** Applies Newton's second law along the slope, using driving force $\frac{64\,000}{16}$, resistance $200 + 16k$ with their k, and a weight component. Condone sign slips.
	$a = 0.602\,\text{m}\,\text{s}^{-2}$	**A1**	
	Total for Question 5	**7**	

Question	Scheme	Marks	Notes
6(a)	Air resistance kv: $1200 = 600 + 15k \Rightarrow k = \ldots$	**M1**	Equates the driving force to 600 plus a resistance proportional to speed at $v = 15$. Condone sign slips.
	$k = 40$	**A1**	
	$P = Fv$ $P = (600 + \text{“}40\text{”} \times 30) \times 30$	**M1**	Takes the driving force at $30\,\text{m}\,\text{s}^{-1}$ as their resistance $600 + 30k$ and multiplies by 30.
	$P = 1800 \times 30 = 54\,000$	**A1 AG**	Correct k and substituted $P = Fv$ leading without error to 54 000.
		[4]	
6(b)	$\frac{54\,000}{20} - (600 + \text{“}40\text{”} \times 20) = 1300a$	**M1**	Uses Newton's second law with driving force $\frac{54\,000}{20}$, their resistance $600 + 20k$ and mass 1300. Condone sign slips.
	$a = 1$	**A1**	
		[2]	
6(c)	$\frac{54\,000}{4} - (600 + \text{“}40\text{”} \times 4) = 1300a$	**M1**	Uses Newton's second law with driving force $\frac{54\,000}{4}$, their resistance $600 + 4k$ and mass 1300. Condone sign slips.
	$a = 9.8$	**A1**	
		[2]	
6(d)	Acceleration at $20\,\text{m}\,\text{s}^{-1}$ reasonable; at $4\,\text{m}\,\text{s}^{-1}$ unrealistically large for a van	**E1**	Interprets both answers, with (c)'s judged too large, for example as close to g. Not for "unrealistic" without saying which.
	Model not valid at low speeds: $F = \frac{P}{v}$ becomes very large as $v \to 0$	**E1**	Concludes the model fails at low speeds, giving the reason that $\frac{P}{v}$ grows as v falls. Accept that full power or tyre grip cannot be sustained then.
		[2]	
	Total for Question 6	**10**	

Question	Scheme	Marks	Notes
7	$k(14)^2 = 1000 \times 9.8 \times 0.08$ [$= 784$]	**M1 A1**	**M1** Equates resistance of the form kv^2 at $v = 14$ to the weight component along the slope, with no driving force.
	$k = 4$	**A1**	
	$P = Fv$ $\frac{P}{10} = 784 + \text{“4”}(10)^2$	**M1 A1**	**M1** Resolves up the slope with zero acceleration, using driving force $\frac{P}{10}$ (or F with $P = 10F$), their kv^2 at $v = 10$ and the weight component. Condone sign slips. **A1** Correct equation oe, such as $F = 1184$.
	$P = 11\,840$	**A1**	awrt 11 800 or 12 000.
	$R = \text{“4”}(12)^2 = 576$	**B1ft**	ft their k, as $144k$.
	Total for Question 7	**7**	

Question	Scheme	Marks	Notes
8	$P = Fv \Rightarrow F = \frac{9000}{12} = 750$	**B1**	
	Whole system, up the slope positive: $750 + 240 + 90 - 900g\sin\alpha = 900a$	**M1 A1**	**M1** Forms an equation of motion for car and trailer with their braking force from $\frac{P}{v}$, both resistances, the weight component and $900a$. Condone sign slips. **A1** Correct equation, unsimplified.
	$a = 0.22$	**A1**	Or -0.22, consistent with their positive direction.
	Trailer, up the slope positive: $T + 90 - 200g\sin\alpha = 200a$	**M1 A1**	**M1** Forms an equation of motion for the trailer, or the car with its 750 N braking force, with the towbar force, its resistance, its weight component and mass times their a. Condone sign slips. **A1** Correct equation, in a or with $a = 0.22$.
	$T = 150$	**A1**	Magnitude 150 N.
	Towbar in compression	**A1**	cso, with the direction of the towbar force on the trailer or car correctly interpreted.
	Total for Question 8	**8**	

Question	Scheme	Marks	Notes
9(a)	$P = Fv \Rightarrow F = \frac{12\,000}{12} = 1000$	**B1**	
	"1000" $= 1000g\sin\theta + 216 + 294$	**M1 A1**	**M1** Forms an equilibrium equation along the slope for the whole system, or one for each body, with driving force, both resistances and a weight component. Condone sign slips. **A1** Correct equation, or a correct pair of equations for the two bodies.
	$\sin\theta = 0.05$	**A1**	
		[4]	
9(b)	$KE_0 = \frac{1}{2} \times 200 \times 14^2 = 19\,600$	**B1**	
	$GPE_0 - GPE_1 = 200gs\sin\theta$ $[= 98s]$	**B1**	GPE lost by the sledge in terms of its distance, $200gs\sin\theta$ or $98s$ oe.
	$19\,600 + 200gs\sin\theta - 294s = 0 \Rightarrow s = \ldots$	**M1 A1**	**M1** Forms a work–energy equation for the sledge with KE, GPE and work against resistance terms. Condone sign slips. Not for Newton's second law with constant-acceleration formulae. **A1** Correct equation oe, with $\sin\theta$ or 0.05.
	$AB = 100\,\text{m}$	**A1**	
		[5]	
	Total for Question 9	**9**	

Question	Scheme	Marks	Notes
10(a)	$F + 5000g\sin\theta - (800 + 500) = 5000(1.2)$ $F + 5000(9.8)(0.06) - 1300 = 6000$	**M1 A1**	**M1** Forms an equation of motion along the slope for the combination, or for each vehicle with a tension, with driving force, weight component and both resistances. Condone sign slips and sin/cos confusion.
	$P = Fv =$ "4360" $\times 14$	**M1**	Uses $P = Fv$ with their driving force and $v = 14$, or substitutes $F = \frac{P}{14}$ into their equation of motion.
	$P = 61\,040$	**A1**	awrt 61 000 oe, such as 61.0 kW.
		[4]	
10(b)	$Pt = 61\,040 \times 7.93$	**B1**	Engine work $7.93P$, with P or their P from (a).
	$GPE_0 - GPE_1 = 5000g\,s\sin\theta = 2940s$	**B1**	GPE lost $5000g\,s\sin\theta$ oe, in terms of the distance.
	$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$ $\frac{1}{2}(5000)(14^2) + 2940s +$ "61 040"$(7.93) - 1300s$ $= \frac{1}{2}(5000)(22^2)$	**M1 A1ft**	**M1** Forms a work–energy equation with the KE change, engine work, GPE change and work against both resistances. Condone sign slips. **A1ft** Correct equation with P or their P from (a).
	$s = \frac{720\,000 - 61\,040(7.93)}{1640}$ $AB = 144\,\text{m}$	**A1**	awrt 144 or 140.
		[5]	
	Total for Question 10	**9**	

Question	Scheme	Marks	Notes
11	$\sin\alpha = \frac{7}{25} \Rightarrow 2500g\sin\alpha = 6860$	**B1**	Correct weight component, soi by $2500g \times \frac{7}{25}$ or 6860 in an equation.
	$\frac{P}{12} - R - 6860 = 2500 \times 2.4$	**M1 A1**	**M1** Resolves along the slope at speed 12 with $\frac{P}{12}$, R, a weight component and 2500×2.4. Condone sign slips and sin/cos confusion.
	$\frac{P}{18} - R - 6860 = 2500 \times 0.4$	**M1 A1**	**M1** Resolves along the slope at speed 18 as at speed 12, with $\frac{P}{18}$ and 2500×0.4.
	$\frac{P}{12} - \frac{P}{18} =$ "12 860" $-$ "7860" $\Rightarrow P = \ldots$	**dM1**	Eliminates P or R from their two equations, each with P divided by the speed, and solves for one unknown.
	$P = 180\,000$	**A1**	Condone an error in the weight component common to both equations, which cancels.
	$R = 2140$	**A1**	
	Total for Question 11	**8**	

Question	Scheme	Marks	Notes
12(a)	$1000P = 2000 \times 20$	**M1**	Uses $P = Fv$ with the driving force equal to the resistance and $v = 20$. Condone a missing or wrong factor of 1000.
	$P = 40$	**A1**	40 000 given as P scores A0.
		[2]	
12(b)	$F = \frac{40\,000}{10} = 4000$	**B1**	
	"4000" $- (220 + 2 \times 10^2) - 2000g \sin\alpha$ $= 2000a$	**M1**	Forms an equation of motion along the slope with their driving force, the refined resistance at $v = 10$, the weight component and $2000a$. Condone sign slips and sin/cos confusion.
	$4000 - 420 - 980 = 2000a$	**A1**	Correct equation, oe.
	$a = 1.3$	**A1**	
		[4]	
12(c)	$F = \frac{40\,000}{U}$	**B1**	Driving force as their power in watts over U.
	"$\frac{40\,000}{U}$" $= 220 + 2U^2 + 2000g \sin\alpha$	**M1**	Equates their driving force, of the form $\frac{k}{U}$, to $220 + 2U^2$ plus the weight component. Condone sign slips and sin/cos confusion.
	$\frac{40\,000}{U} = 1200 + 2U^2$	**A1**	Correct equation, oe.
	$U^3 + 600U - 20\,000 = 0 \Rightarrow U = \ldots$	**dM1**	Multiplies through by U to form a cubic and solves it by any means, including a calculator.
	$U = 20$	**A1**	Other roots need not be considered.
		[5]	
	Total for Question 12	**11**	